Module 14)

python – collection , function and modules

1.Accessing List

1. Understanding how to create and access elements in a list.

- List is a collection data type in python. We can store multiple values single variable it is called list.
- Denoted by [ ]
- List is mutable.
- List is orderable.
- List allow duplicate values.

- **Creating a List**
  - You can create a list by placing items (elements) inside square brackets [ ], separated by commas:
  - fruits = ["apple", "banana", "cherry"]

- **Accessing Elements in a List**

  1. Access by Index

  Python uses zero-based indexing, which means the first element is at index 0 .

2. **Access from the End (Negative Indexing)**
   You can also use negative numbers to access elements from the end of the list .

3. **Slicing a List**
   You can get a sublist using slicing .

2. Indexing in lists (positive and negative indexing).

- A list in Python is an ordered collection of items. You can access individual items using indexes.

- ### Positive Indexing

  Indexing starts at 0.
  The first element has index 0, the second has index 1, and so on.

  Example :

  ```
  fruits = ["apple", "banana", "cherry", "date"]

  print(fruits[0])  # Output: apple
  print(fruits[1])  # Output: banana
  print(fruits[2])  # Output: cherry
  print(fruits[3])  # Output: date
  ```

- ### Negative Indexing

  Negative indexes count from the end of the list.
  -1 is the last item, -2 is the second last, etc..

Example

```
fruits = ["apple", "banana", "cherry", "date"]

print(fruits[-1])  # Output: date
print(fruits[-2])  # Output: cherry
print(fruits[-3])  # Output: banana
print(fruits[-4])  # Output: apple
```

3. Slicing a list: accessing a range of elements.

- Syntax
  list[start:stop]

  - start: index to begin the slice
  - stop: index to end the slice

- Example

```
fruits = ["apple", "banana", "cherry", "date", "elderberry"]

print(fruits[1:4])   # Output: ['banana', 'cherry', 'date']
print(fruits[0:3])   # Output: ['apple', 'banana', 'cherry']

print(fruits[-3:-1])  # Output: ['cherry', 'date']
```